

NExT

A Schlumberger Company

Map-based volume calculations and final challenge



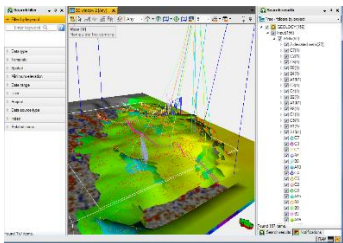
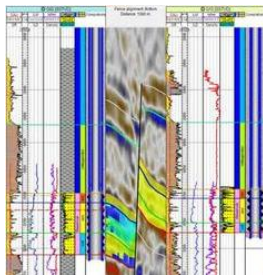
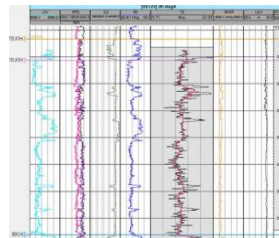
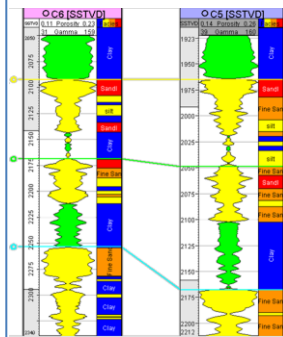
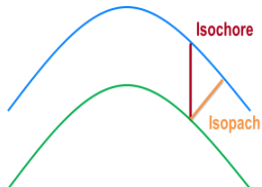
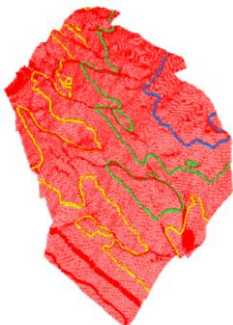
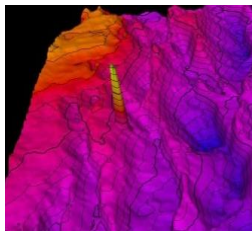
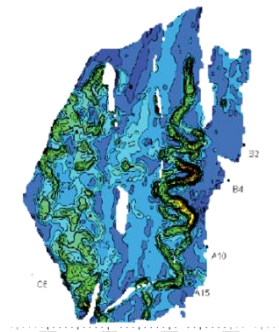
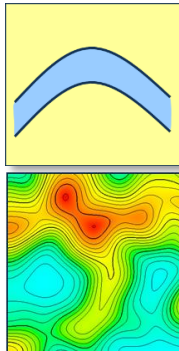
Schlumberger-Private

Learning objectives

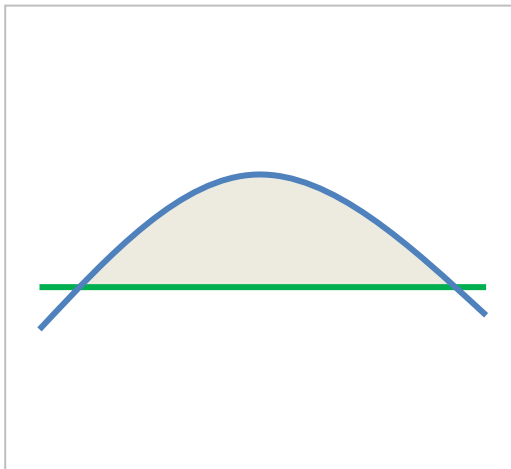
In this module, you learn how to:

- use the Map-based volume calculation
- perform a final challenge

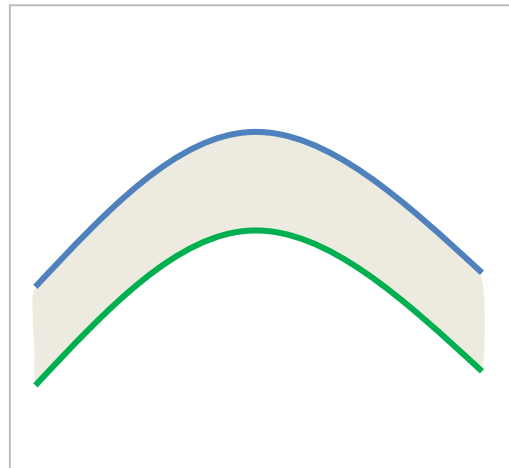
Petrel Geology workflow

Introduction			Find your data	Data display in Well section window	Log editing	Well correlation
About this manual	Challenge	Agenda				
Isochore and Isopach processing			Gridding and quality improvement	Surface editing and operations	Log property mapping	Map-based volume calculations
						

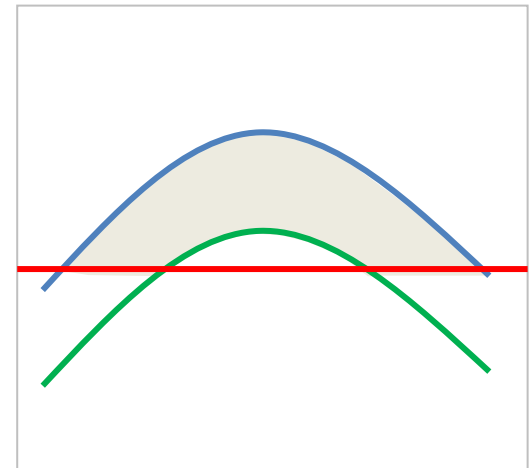
Volume calculations



Surface and constant/plan



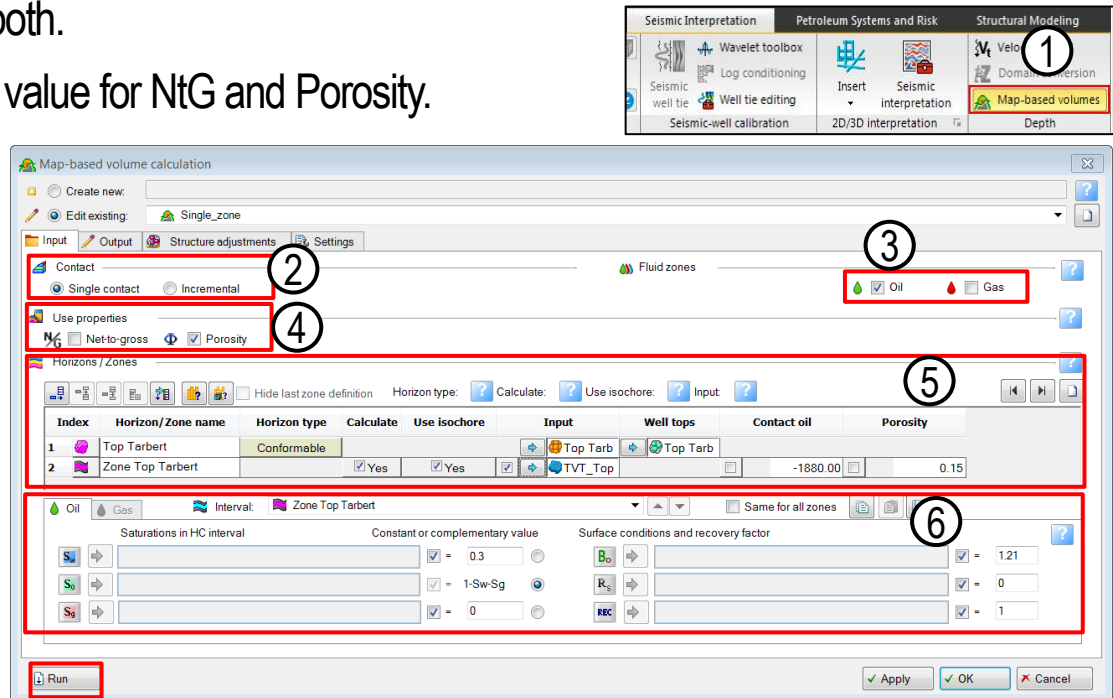
Surface - Surface



Volume/area vs. depth

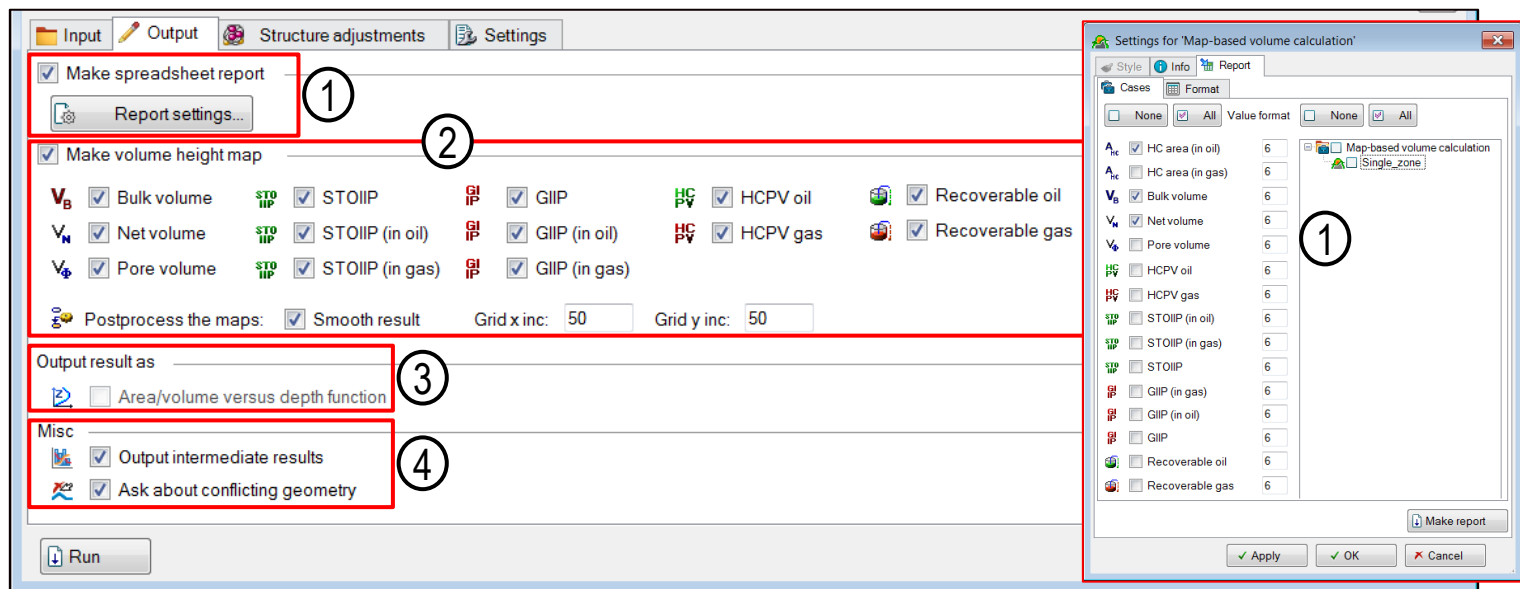
Map-based volume calculation dialog box: Input tab

1. Access **Map-based volumes** from the **Depth** group on the **Seismic interpretation** tab.
2. On the **Input** tab, set a surface or a constant value as a contact. If you use *Incremental*, insert the minimum, maximum and increment values.
3. Calculate volumes for Oil or Gas or both.
4. (Optional) Use a surface or constant value for NtG and Porosity.
5. Set the scenario for the calculation.
6. Set the Saturations to be used for each zone.



Map-based volume calculation dialog box: Output tab

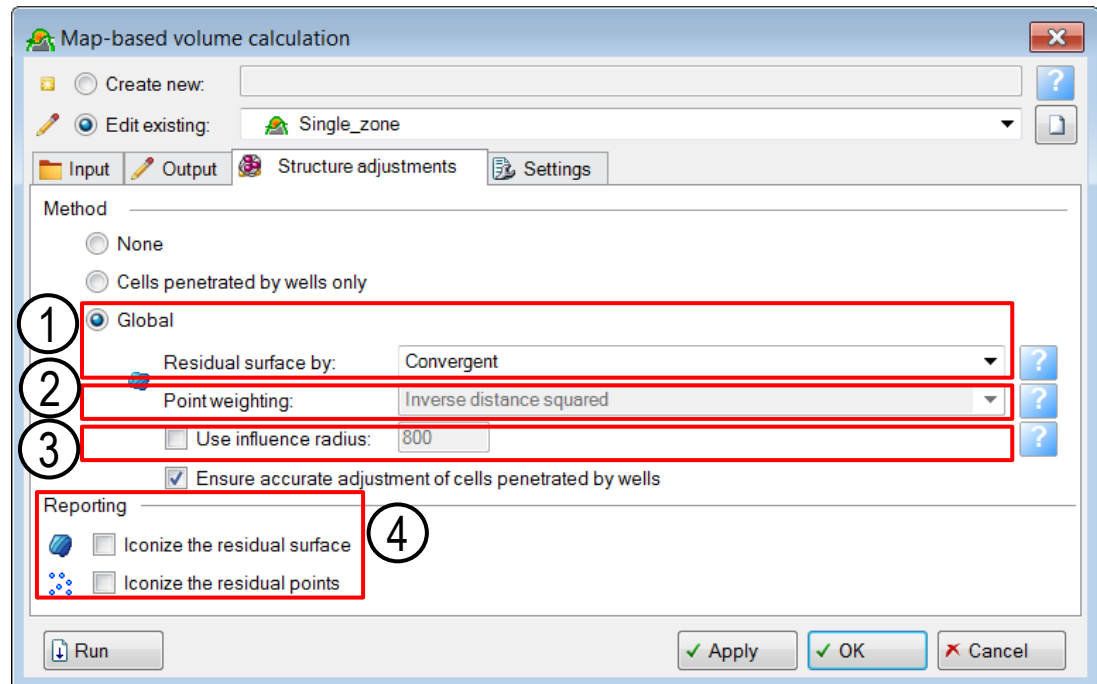
1. A detailed, user-specified report can be made any time after running the calculation.
2. Select volume properties to be made. These represent the property value per unit area.
3. Output the result as an area/depth versus volume function for all individual zones.
4. (Optional) Select to output intermediate results and to be asked about conflicting geometry.



Map-based volume calculation dialog box: Structural adjustments tab

You can choose to have structural adjustments based on well tops in the **Input** tab.

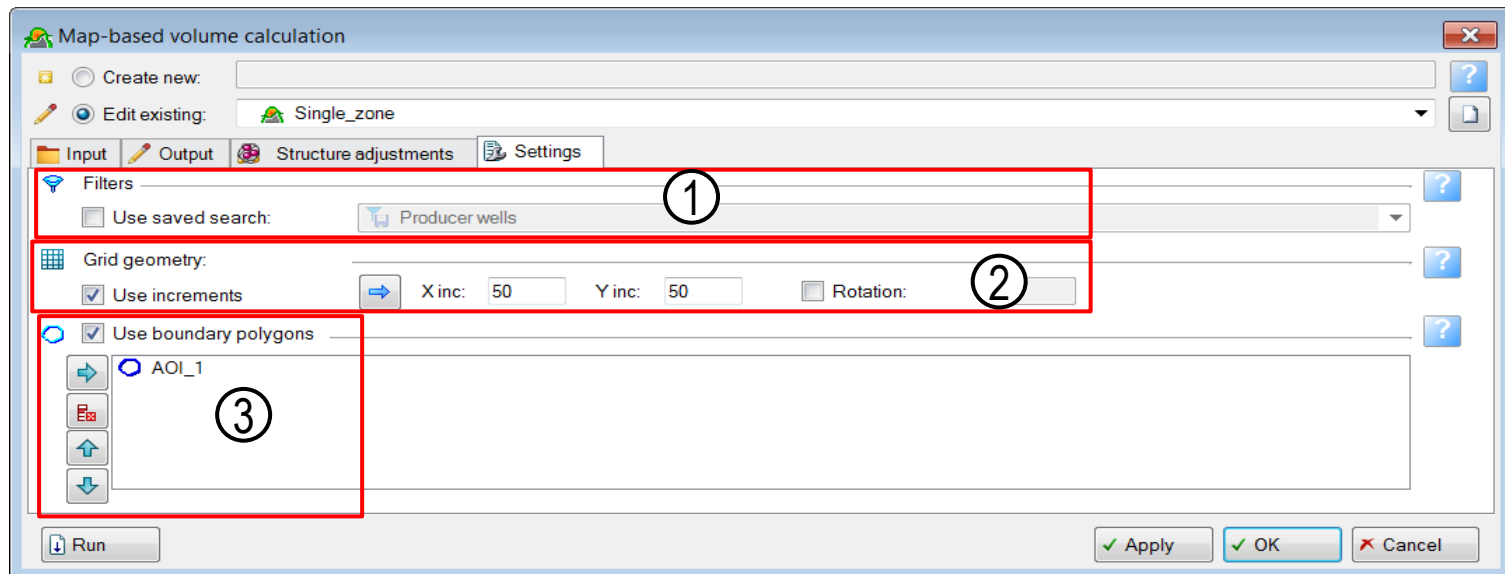
1. Select Global adjustment to choose the algorithm that will calculate the residual surface: Convergent, Minimum curvature, Moving average, Cos expansion.
2. Use Point weighting with some algorithms for the distance of the node from the well.
3. Set Influence radius to limit the structural adjustment around the well.
4. Iconize the residual surface or points to more easily inspect the changes.



Map-based volume calculation dialog box: Settings tab

Define the settings for the volume maps created.

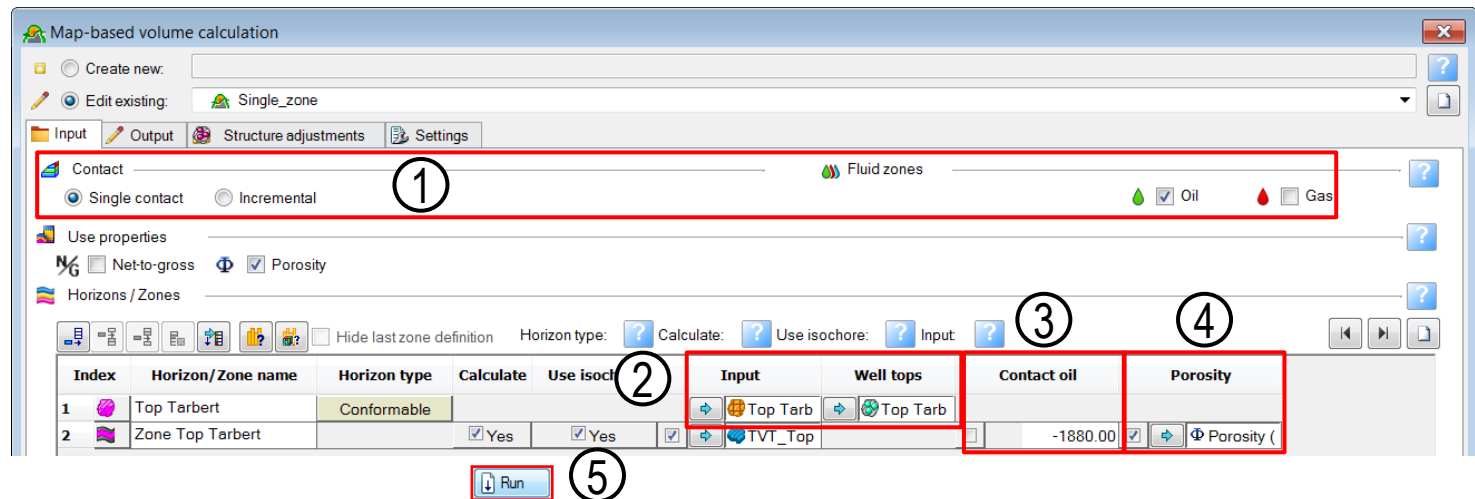
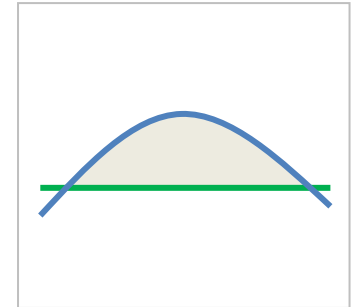
1. Filters: Use a saved search
2. Manually set the grid geometry to control the map resolution (XY increment and rotation)
3. Select a boundary polygon to define the limits of the maps created.



Volume between a surface and a contact/plane

To calculate the volume between a surface and a flat contact:

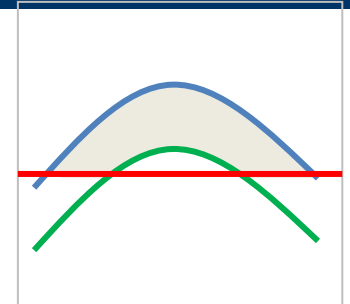
1. Select Single contact and which fluid phases are present.
2. Drop in the top surface to be used.
3. Specify the depth of the contact to be used.
4. (Optional) Use Porosity or NtG as surfaces or constant values.
5. Click *Run*.



Volumes between two surfaces and a constant

To calculate the volume between two surfaces and a flat contact:

1. Select Single contact and which fluid phases are present.
2. Select the *Hide last zone definition* check box.
3. Drop in the surfaces to be used
4. Specify the depth of the contact to be used
5. (Optional) Use Porosity or NtG as surfaces or constant values.
6. Click *Run*.



Map-based volume calculation

Create new:

Edit existing:

Input Output Structure adjustments Settings

Contact ☒ Single contact ☐ Incremental ①

Fluid zones ☒ Oil ☐ Gas

Use properties

☒ Net-to-gross ☒ Porosity

Horizons / Zones ②

☒ Hide last zone definition ② Horizon type: Calculate: Use isochore: Input ③ ④

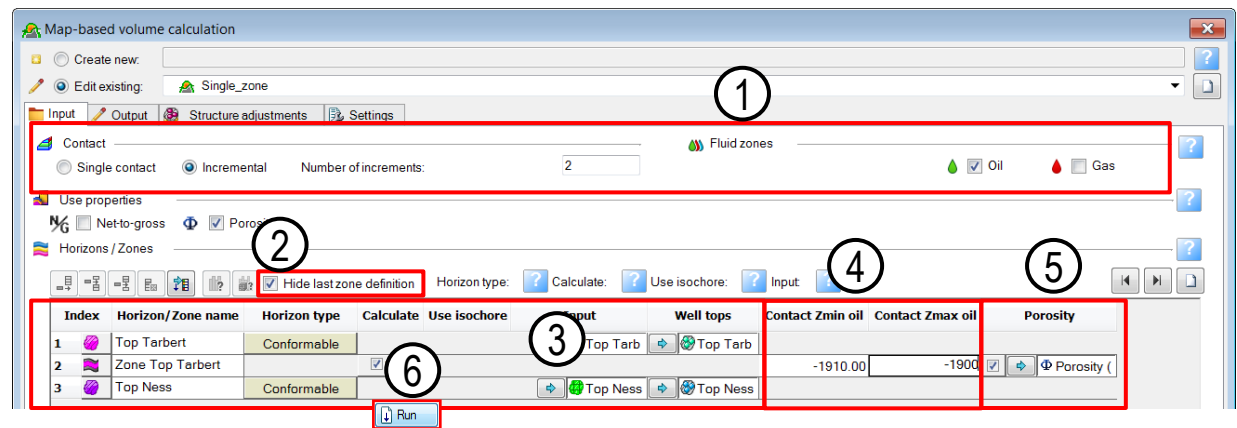
Index	Horizon/Zone name	Horizon type	Calculate	Use isochore	Input	Well tops	Contact oil	Porosity
1	Top Tarbert	Conformable			Top Tarb	Top Tarb		
2	Zone Top Tarbert		☒ Yes				-1880.00	☒ Φ Porosity (
3	Top Ness	Conformable			Top Ness	Top Ness		

⑤

Incremental volumes between two surfaces

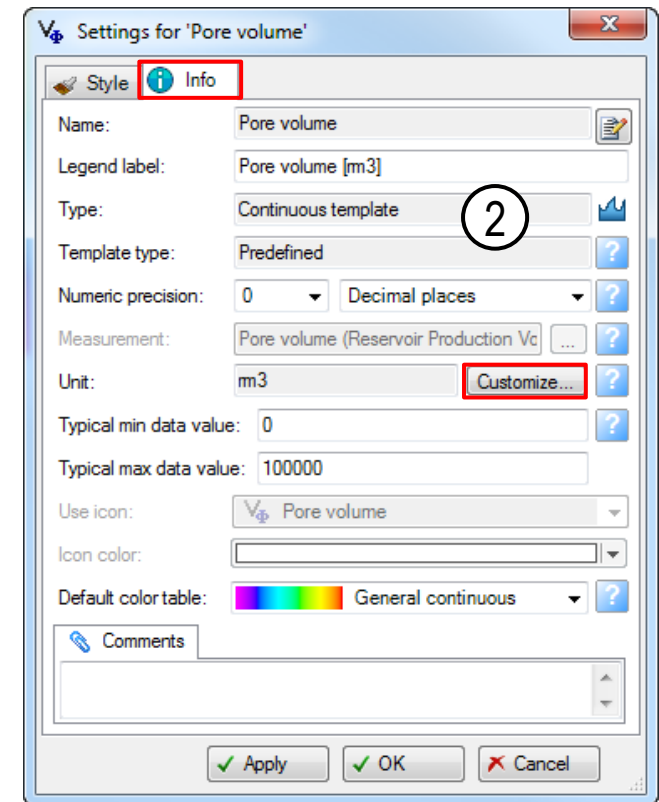
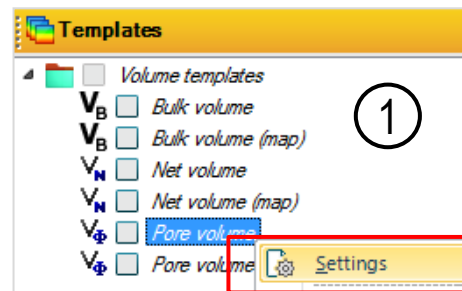
To calculate the volume between two surfaces and a flat contact:

1. Select Incremental contact, set the number of increments, and specify fluid phases are present.
2. Select the *Hide last zone definition* check box.
3. Drop in the surfaces to be used.
4. Specify the the minimum and maximum contact depth.
5. (Optional) Use Porosity or NtG as surfaces or constant values.
6. Click *Run*.



Change volume units in a report (1)

1. In the **Templates** pane, expand the *Volume templates* folder and double-click any template to open the **Settings** dialog box for that template.
2. Open the **Info** tab and click *Customize*.
A new **Settings** dialog box opens with a unit list from which you select a unit type, for example, bbl.



Change volume units in a report (2)

3. Close the **Settings** dialog boxes.
4. Right-click the case you just created and click *Make map-based volumetrics report*.
5. View the report with the new volume units.

The screenshot displays the 'Settings for Data_repository_Tsacco.pet' dialog box. The 'Measurement:' list on the left has 'Reservoir Production Volume' selected. The 'Unit:' list on the right has 'm3' selected. The 'Templates using Reservoir Production Volume:' list on the right includes 'Pore volume [m3]', 'HCPV oil [m3]', 'HCPV gas [m3]', 'Rock volume [m3]', and 'Reservoir production volume [m3]'. The 'Close' button is highlighted with a red box.

The 'Cases' panel on the left shows a list of cases, with 'Case' selected. The 'Make map-based volumetrics report' option is highlighted with a red box.

The 'Make map-based volumetrics report' context menu is shown, with the 'Make map-based volumetrics report' option highlighted with a red box.

The resulting report table is shown below:

	A	B	C	D
1	Petrel 2014.1 (64-bit) beta	Schlumberger		
2	User name	TSacco		
3	Date	Wednesday, May 28 2014 09:01:04		
4	Project	Data_repository_Tsacco.pet		
5				
6	Input XY unit	m		
7	Input Z unit	m		
8				
9	Input data			
10	Zone(s)	Top surface	Base surface/sochore	Contact
11	Zone Top Tarbert depth	Top Tarbert depth	Top Ness depth	-2000.00
12				
13	Boundaries			
14	Boundary polygon			
15				
16	Case	HC Area[10^6 m2]	Bulk volume[10^6 m3]	
17	Case	34.65	1539	
18				
19	Totals all result types			
20				
21	Zones	HC Area[10^6 m2]	Bulk volume[10^6 m3]	
22	Zone Top Tarbert depth	34.65	1539	
23				
24	Detailed results			
25				
26	Zones	HC Area[10^6 m2]	Bulk volume[10^6 m3]	
27	Zone Top Tarbert depth	34.65	1539	

Summary

In this module, you learned about:

- the **Map-based volume calculation** tool
- the three basic volume calculation scenarios
- how and when to use single and incremental contacts

Exercises

Perform Map-based volume calculations.